AWS GLOBAL ACCELERATOR

WHAT IS AWS GLOBAL ACCELERATOR

AWS Global Accelerator is a global networking service that routes users to the nearest healthy AWS endpoint using the AWS global network. It improves application performance, provides static IPs, and enables automatic cross-region failover.

SCENARIOS

Without Global Accelerator
User (Germany) ----> Public Internet ----> Mumbai ALB ----> Mumbai EC2

Problems:

1. Higher latency
2. Internet congestion
3. Slow failover
4. DNS caching issues

With Global Accelerator
User (Germany) ----> Nearest AWS Edge Location ----> AWS Global Network ----> Frankfurt ALB ----> Frankfurt EC2

Benefits:

1. Lower latency
2. Faster response time
3. Automatic failover
4. Uses AWS backbone network
5. Single global endpoint

WHY DO WE USE GLOBAL ACCELERATOR?

1. Reduce Latency
Suppose your application is deployed in:

1. Mumbai
2. Frankfurt

If a German user accesses the application:
Without GA

1. Germany --> Internet --> Mumbai
2. Response is slower.

With GA

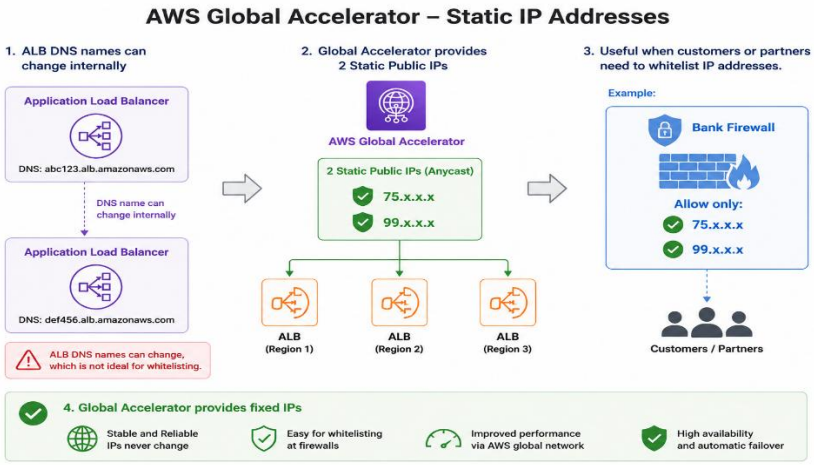
1. Germany --> Frankfurt
2. Response is faster because traffic enters AWS through the nearest edge location.

2. Automatic Disaster Recovery

1. Suppose Frankfurt region goes down.
   Frankfurt ALB
2. Global Accelerator automatically sends traffic to:
   Mumbai ALB
3. Users don't need to change DNS.

3. Static IP Address

1. ALB DNS names can change internally.
2. Global Accelerator provides:
   1. 2 Static Public Ips
   2. Useful when customers or partners need to whitelist IP addresses.
   3. Example:



   4. Global Accelerator provides fixed IPs.

INDUSTRY USE CASES

1. Netflix-like Streaming Platform

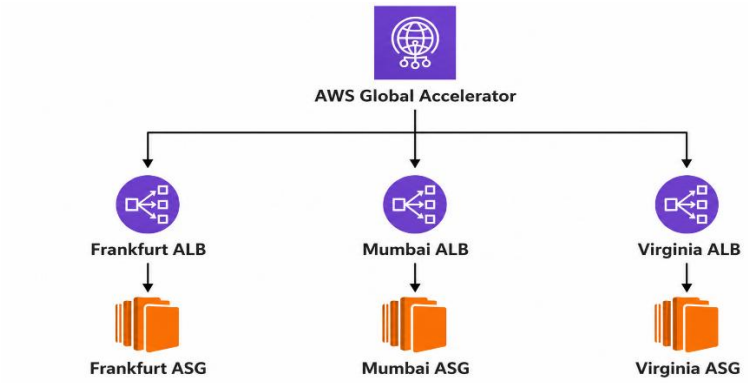
1. Users worldwide access:
   1. India → Mumbai
   2. Germany → Frankfurt
   3. USA → Virginia
2. Global Accelerator routes users to the nearest healthy region.

2. Banking Applications

1. Banks often whitelist IPs.
2. ALB cannot provide fixed IPs.
3. Global Accelerator solves this.

3. Gaming Applications

1. Games need very low latency.
2. Example:
    1. PUBG
    2. Fortnite
    3. Call of Duty
3. Global Accelerator routes players to the nearest game server.

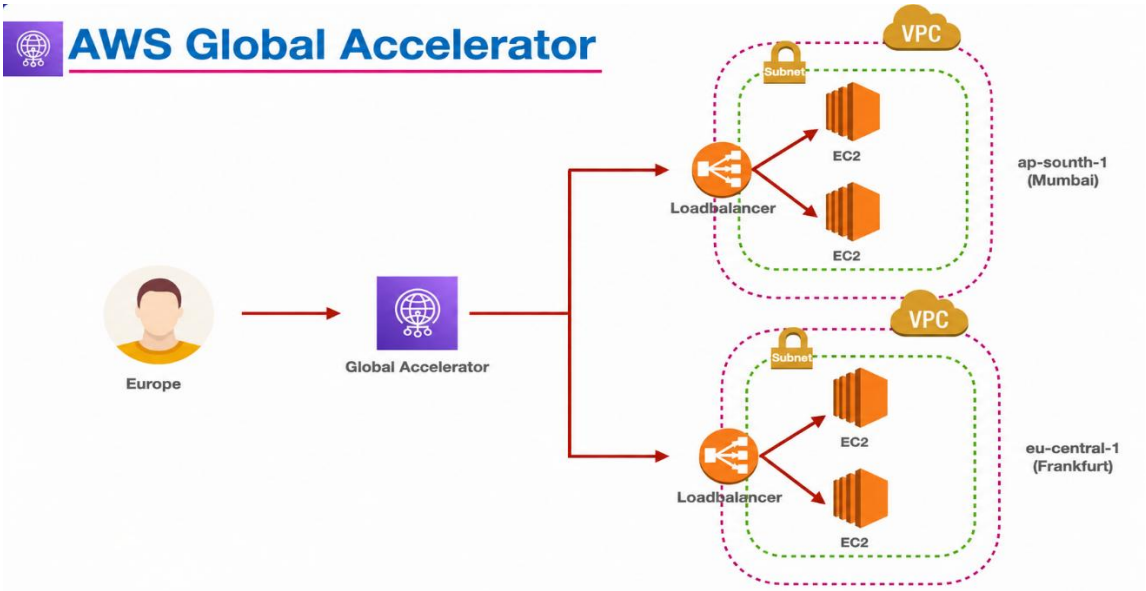


WHEN SHOULD WE USE IT?

Use Global Accelerator when:

1. Application is deployed in multiple AWS regions.
2. Low latency is required.
3. Static public IPs are needed.
4. Fast disaster recovery is required.
5. Users are spread across different countries.

PRACTICAL ON AWS GLOBAL ACCELERATOR



Mumbai

1. create a vpc (10.0.0.0./16)
2. create a subnet
    1. public-subnet-1(10.0.0.0/24)-1a
    2. public-subnet-2(10.0.1.0/24)-1b
3. create a internet gateway and attach to vpc
4. create a public-route table
    1. route ----- internet gateway
    2. subnet --- public-subnet 1&2
5. create a ec2
    1. apt update
    2. apt install apache2 -y
    3. echo "<h1> Hello From Mumbai </h1>" > /var/www/html/index.html
    4. systemctl restart apache2

6. create a target group
    1. name ---- mumbai-tg
    2. vpc ----- Mumbai vpc
    3. path ---- /
    4. register -- mumabi-ec2
    5. create a target group

7. create a load balancer
    1. Name ----- mumbai-alb
    2. vpc -------- mumbai-vpc

3. AZ --------- public-subnet-1&2
4. forward -- mumbai-tg
5. create a load balancer

Frankfurt

1. create a vpc (12.0.0.0./16)
2. create a subnet
    1. public-subnet-1(12.0.0.0/24)-1a
    2. public-subnet-2(12.0.1.0/24)-1b
3. create a internet gateway and attach to vpc
4. create a public-route table
    1. route ----- internet gateway
    2. subnet --- public-subnet 1&2
5. create a ec2
    1. apt update
    2. apt install apache2 -y
    3. echo "<h1> Hello From Frankfurt </h1>" > /var/www/html/index.html
    4. systemctl restart apache2

6. create a target group
    1. name ---- frankfurt-tg
    2. vpc ------- frankfurt-vpc
    3. path ----- /
    4. register -- frankfurt-ec2
    5. create a target group

7. create a load balancer
    1. name ----- frankfurt-alb
    2. vpc -------- frankfurt-vpc
    3. AZ --------- public-subnet-1&2
    4. forward -- frankfurt-tg
    5. create a load balancer

AWS GLOBAL ACCELERATOR

1. search global accelerator
2. create accelerator
3. name ------- my-accelerator
4. type --------- standard
5. ports -------- 80
6. protocol ---- TCP

7. affinity ------ none
8. endpoint
    1. region ----- Mumbai
    2. traffic ------ 100
9. add another endpoint
    1. region ----- Frankfurt
    2. traffic ----- 100
10. listener:80 TCP
    1. endpoint group: Mumbai
        1. endpoint type ---- application load balancer
        2. endpoint ---------- (mumbai-alb)
    2. endpoint group: Frankfurt
        1. endpoint type ---- application load balancer
        2. endpoint ---------- (frankfurt-alb)
11. create global accelerator
12. check status of global accelerator (wait atleast 5min)
13. copy and paste the dns of global accelerator

NOTE:

1. If I am accessing the application from Europe, my request should be routed to the Frankfurt region.
2. Even if I refresh the URL multiple times, I should continue to receive the Frankfurt response.
3. This is because Frankfurt is the nearest healthy AWS region and provides the lowest latency to the user.
4. If the Frankfurt endpoint becomes unhealthy, Global Accelerator will automatically route traffic to the Mumbai region.
